Kingdom of Saudi Arabia

Majmaah University

College of computer and information sciences

Assignment 2

CS 473

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Assignment 1- CS 470 (Submission Date- within 1 week)

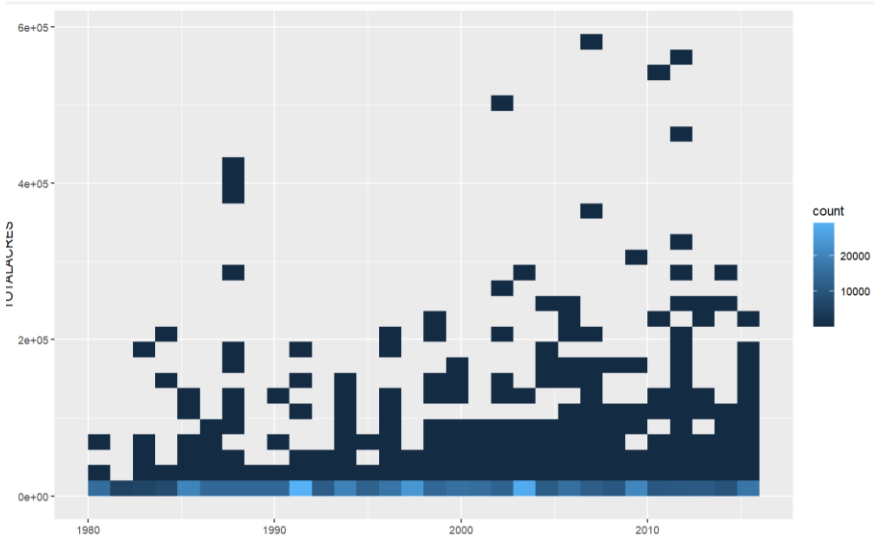
1. Create 2D bin and hex charts for StudyArea.csv file.

Codes:

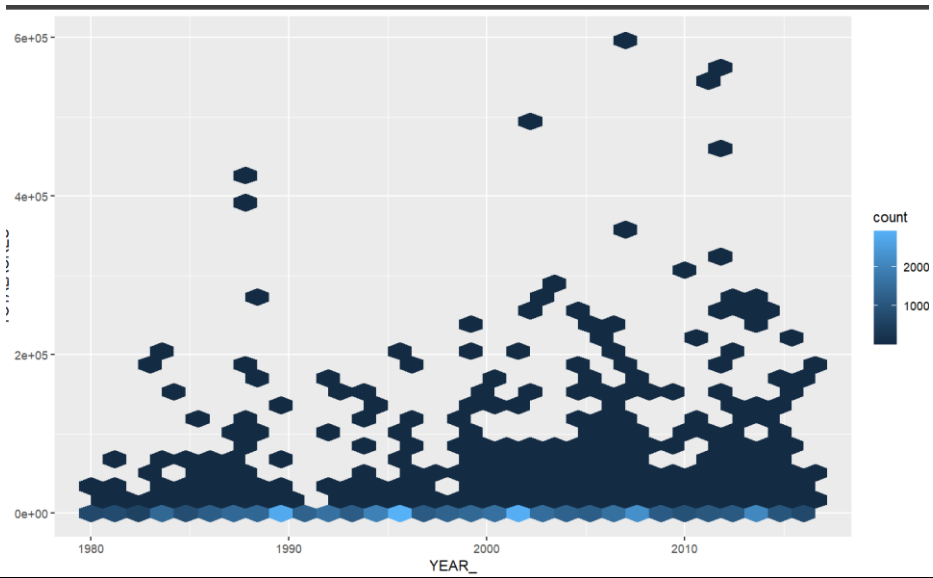
```
#step1 ;load library
library(tidyverse)

# 2D bin
dff <- read_csv("StudyArea.csv", col_types = list(UNIT = col_character()), col_names = TRUE)
ggplot(data = dff) + geom_bin2d(mapping = aes(x=YEAR_,y=TOTALACRES))

# hex chart
ggplot(data=dff) + geom_hex(mapping = aes(x=YEAR_, y=TOTALACRES))
```

Output:

2D bin



Hex chart



2. How you can generate summary statistics for any data, you can use StudyArea.csv file, also submit screenshots from your system.

Codes:

```
#step1 ;load library
library(tidyverse)

#step2 ; load data set and create sub column
dfw <- read_csv("StudyArea.csv", col_types = list(UNIT = col_character()), col_names = TRUE)

df <- select(dfw, ORGANIZATI, STATE, YEAR_, TOTALACRES, CAUSE)
grp <- group_by(df, YEAR_)
sm <- summarize(grp, totalacres = sum(TOTALACRES))

#step3: summary statistics
summary(data)
```

Output:

```
   STATE_FIPS      TOTALACRES
 Min.   : 4.0   Min.   :     0.0
 1st Qu.: 6.0   1st Qu.:     0.1
 Median :16.0   Median :     0.1
 Mean   :22.2   Mean   :   191.1
 3rd Qu.:41.0   3rd Qu.:     1.0
 Max.   :56.0   Max.   :590620.0
```

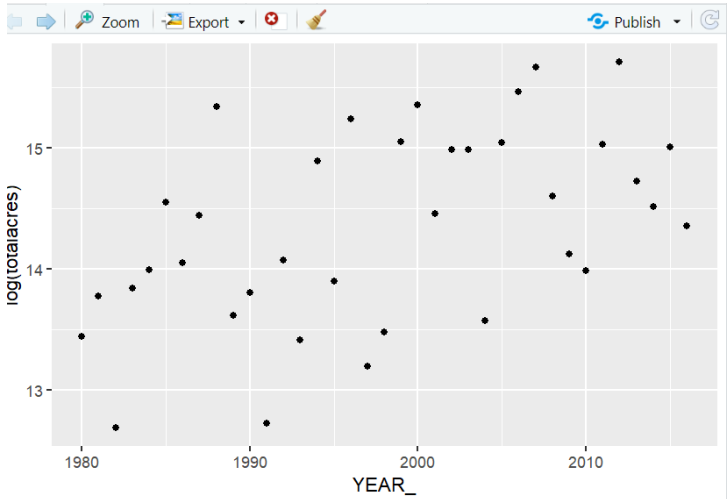
3. Write the steps with screenshots of output to create scatterplot having regression line for StudyArea.csv data.

Codes:

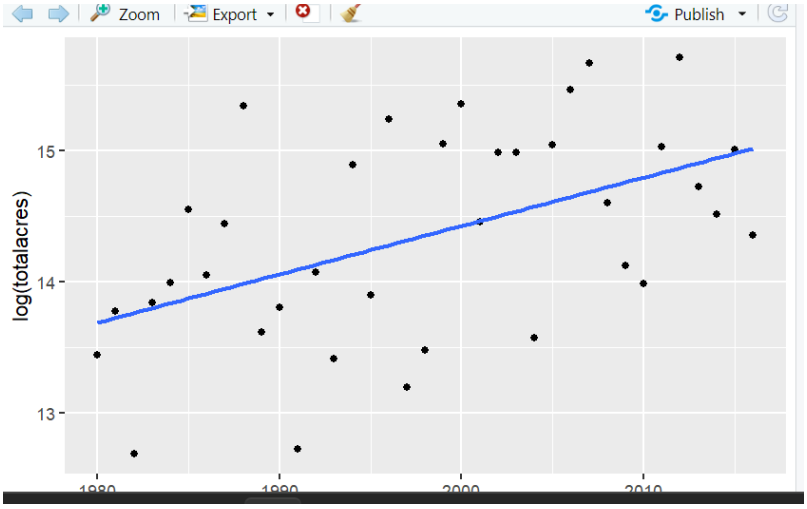
```
#step1 ;load library
library(tidyverse)

#step2 ; load data set and create sub column
dfw <- read_csv("StudyArea.csv", col_types = list(UNIT = col_character()), col_names = TRUE)

df <- select(dfw, ORGANIZATI, STATE, YEAR_, TOTALACRES, CAUSE)
grp <- group_by(df, YEAR_)
sm <- summarize(grp, totalacres = sum(TOTALACRES))

#step3: scatterplot
ggplot(data=sm) + geom_point(mapping = aes(x=YEAR_, y=log(totalacres)))
#step4: regression line to the scatterplot
ggplot(data=sm, aes(x=YEAR_, y=log(totalacres))) + geom_point() + geom_smooth(method=lm, se=FALSE)

#step5:
ggplot(data=sm, aes(x=YEAR_, y=log(totalacres))) + geom_point() + geom_smooth(method=loess, se=TRUE)
```

Output:

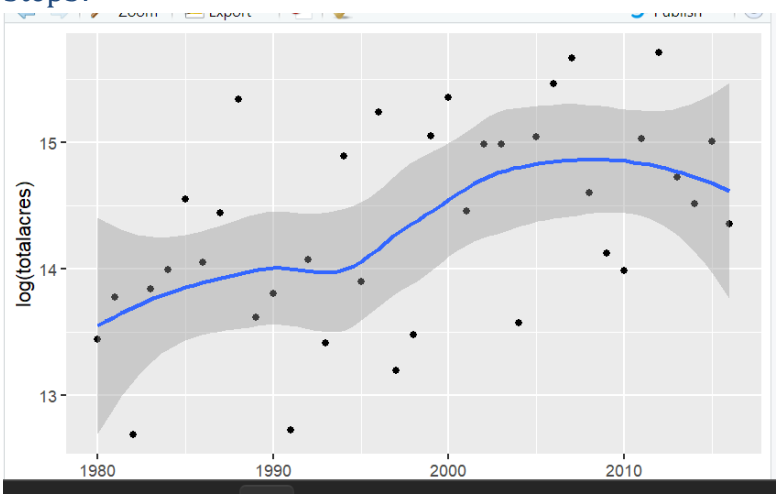
Step3:



Step4:



Step5:



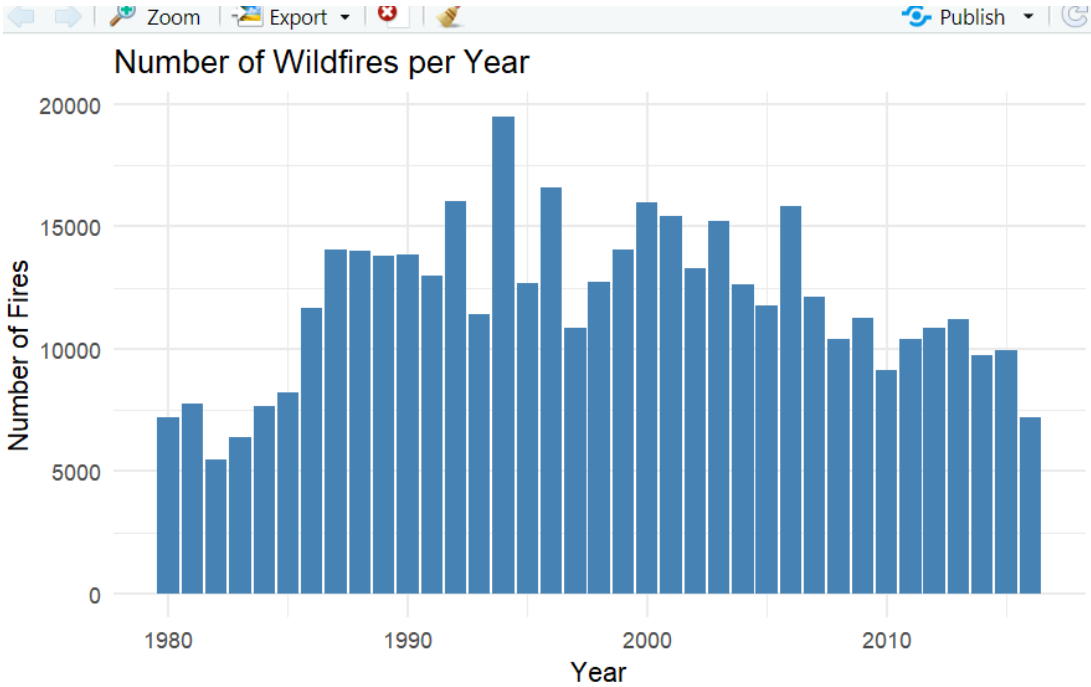
4. How you can create the bar charts and violin plots using StudyArea.csv data

- **bar charts**

Codes:

```
library(tidyverse)

# Load the data
df <- read_csv("StudyArea.csv")

# Create bar chart: Number of fires per year
df %>%
  count(YEAR_) %>%
  ggplot(aes(x = YEAR_, y = n)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Number of Wildfires per Year", x = "Year", y = "Number of Fires") +
  theme_minimal()
```

Output:

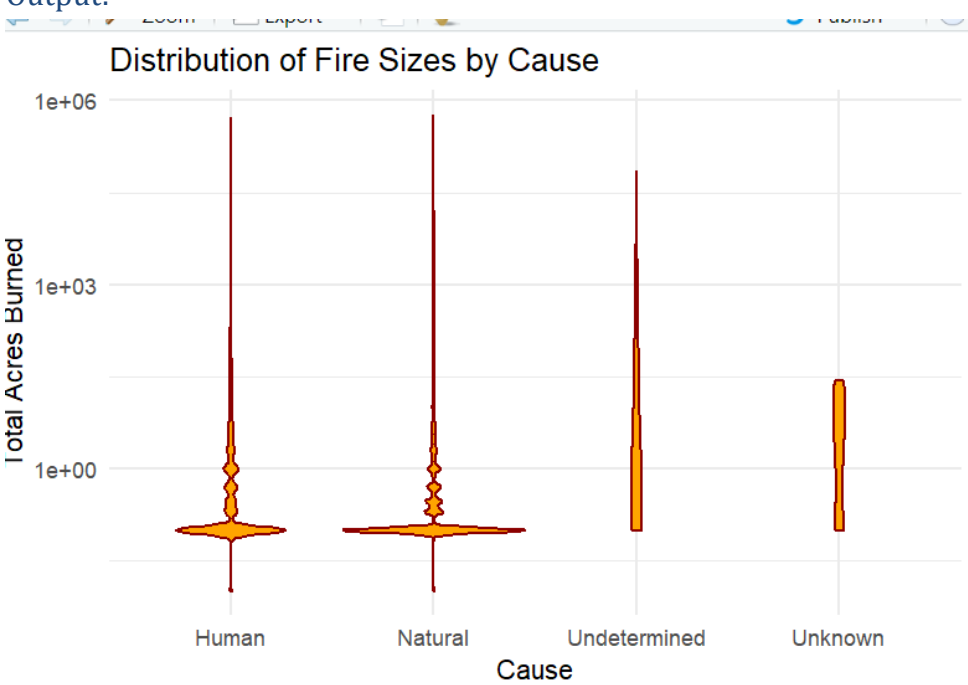


- **violin plots:**

Code:

```
library(tidyverse)

# Load the data
df <- read_csv("StudyArea.csv")

df %>%
  filter(!is.na(CAUSE), TOTALACRES > 0) %>%
  ggplot(aes(x = CAUSE, y = TOTALACRES)) +
  geom_violin(fill = "orange", color = "darkred") +
  scale_y_log10() +  # Log scale to handle skewed data
  labs(title = "Distribution of Fire Sizes by Cause", x = "Cause", y = "Total Acres Burned") +
  theme_minimal()
```

Output:



5. write the steps to create a basemap using ggmap.

Steps to create a basemap:

1. Install and load ggmap
2. Get your Google Maps API key (if using Google maps)
3. Define your area of interest
4. Download the basemap
5. Plot your data on it

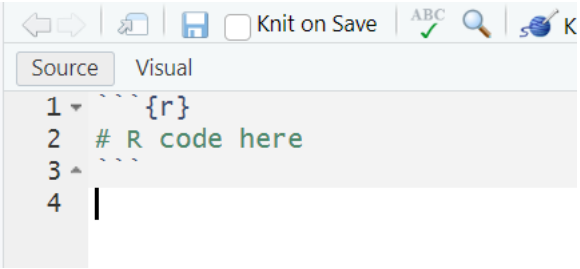
Example code:

```
# Load required libraries
library(ggmap)
library(tidyverse)

# Set location and get the map
myLocation <- "California"
map <- get_stamenmap( bbox = c(left = 110, bottom = -40, right =160, top = -10), zoom = 4, maptype = "watercolor")

#
ggmap(map) + theme_void() + theme( plot.title = element_text(colour = "orange"), panel.border = element_rect(colour = "grey", fill=NA, size=2) )
myMap <- get_map(location = myLocation, zoom = 6)
dfWildfires <-
read_csv("StudyArea_SmallFile.csv", col_names = TRUE)
df <- select(dfWildfires, STATE, YEAR_, TOTALACRES, DLATITUDE, DLONGITUDE)
df <- filter(df, TOTALACRES >= 1000 & STATE == 'California')
ggmap(myMap) + geom_point(data=df, aes(x = DLONGITUDE, y = DLATITUDE))
```

6. Write the steps to create R markdown file.

1. Open RStudio.
2. Go to `File` → `New File` → `R Markdown`.
3. In the dialog box that appears:
   - Select Document (this is the default).
   - Enter a title, for example: *"Creating Maps with R"*.
   - Enter the author name.
   - Select the output format (e.g., PDF, HTML, Word).
4. Click OK.
   RStudio will create a new `.Rmd` file that contains:
   - A YAML header (at the top, surrounded by `---`).
   - Some default text and code chunks.
5. Edit the file:
   - You can add or remove code/text as needed.
   - Remove everything except the header if starting from scratch.
6. Save the file:
   - Go to `File` → `Save As...` and save it with a name .
   -
7. Demonstrate the use of code chunk and header options with Mapsetup context.

   code chunk is a section of R code that is embedded in an R Markdown file. It's defined using three backticks (```) followed by {r}



Example with Chunk Name and MapSetup

```
ource   Visual
1  ```{r MapSetup, warning=FALSE, error=FALSE, message=FALSE}
2  # Load necessary libraries
3  library(ggplot2)
4  library(ggmap)
5  library(readr)
6  library(dplyr)
7
8  # Set location and get base map
9  myLocation <- "California"
10 myMap <- get_map(location = myLocation, zoom = 6)
11 ggmap(myMap)
12 ```
13
```

8. Have the number of wildfires increased or decreased in the past few decades for StudyArea data, shown all steps with visualization?

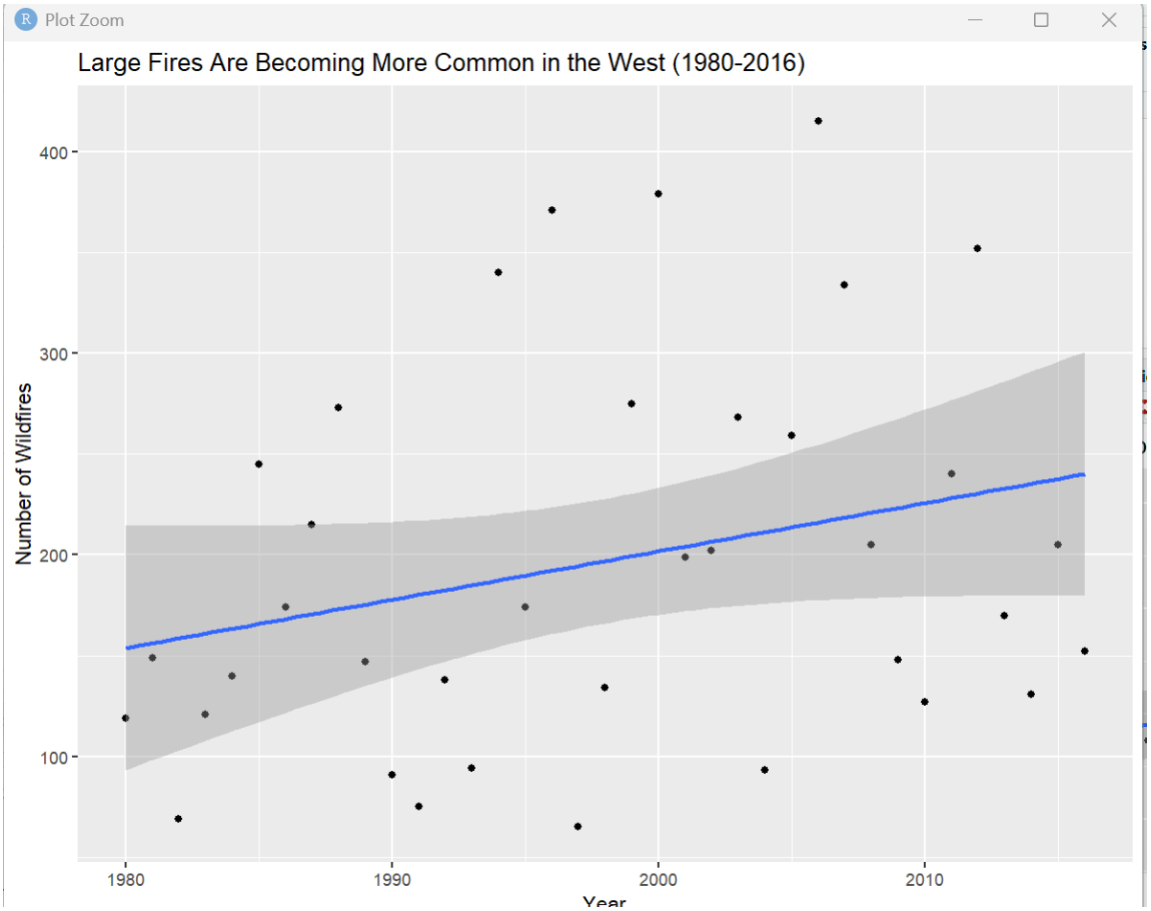
Analysis Steps:

1. Filter the data to include only large wildfires (>1,000 acres)
2. Group by year and count the number of fires
3. Visualize with a scatterplot and regression line
4. Perform statistical analysis

Codes:

```
library(tidyverse)


# Load and prepare data
df <- read_csv("StudyArea.csv", col_types = list(UNIT = col_character()))
fire_counts <- df %>%
  select(STATE, YR = YEAR_, ACRES = TOTALACRES) %>%
  filter(ACRES >= 1000) %>%
  group_by(YR) %>%
  summarize(count = n())

# Visualization
ggplot(fire_counts, aes(x = YR, y = count)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  ggtitle("Large Fires Are Becoming More Common in the West (1980-2016)") +
  xlab("Year") +
  ylab("Number of Wildfires")

# Statistical test
model <- lm(count ~ YR, data = fire_counts)
summary(model)
```

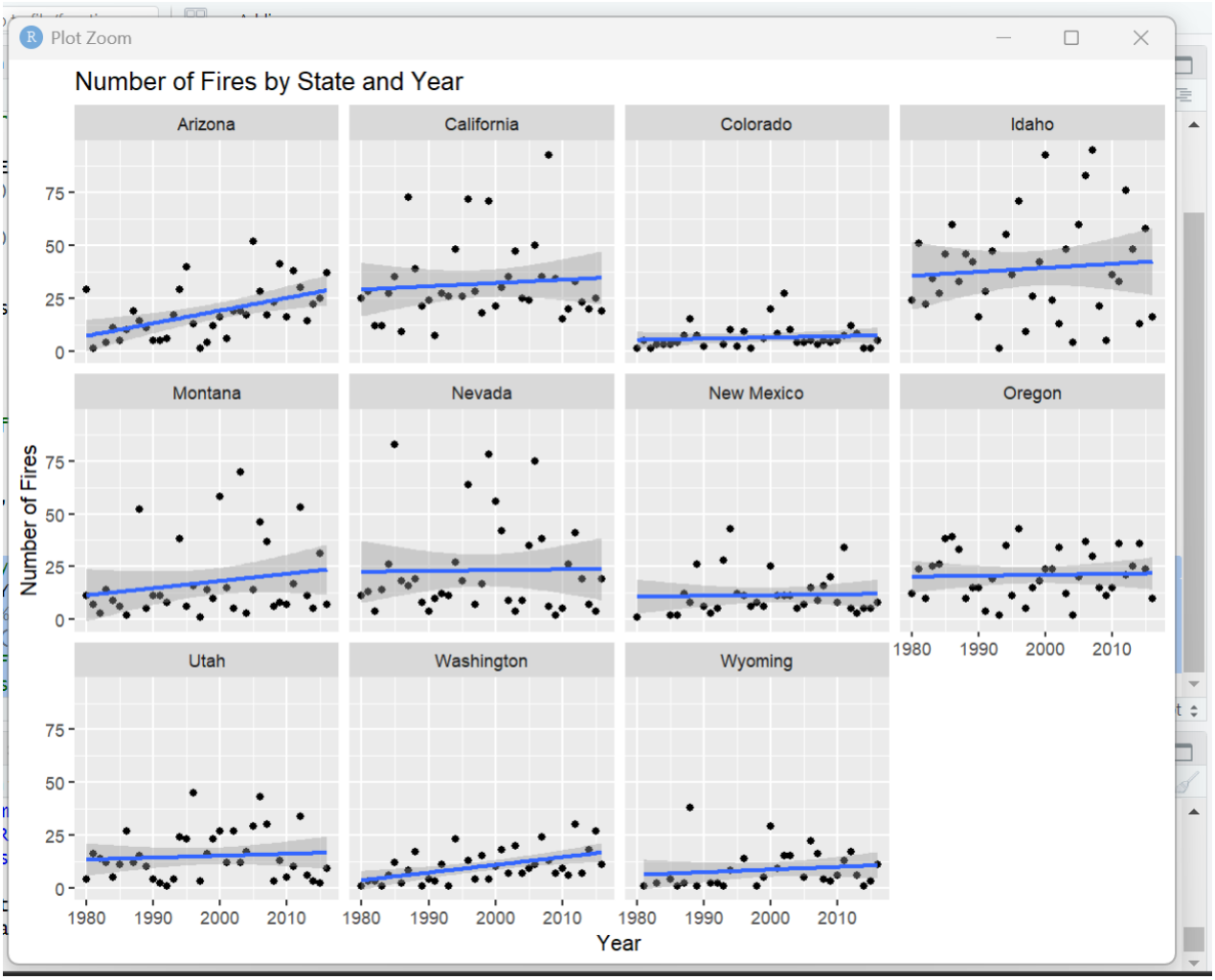
Output:



Codes:

```
dff <- read_csv("StudyArea.csv", col_types = list(UNIT = col_character()), col_names = TRUE)
dff %>%select(STATE, YR = YEAR_, ACRES = TOTALACRES, CAUSE) %>% filter(ACRES >= 1000)%>%
group_by(STATE, YR) %>%summarize(cnt = n()) %>%
  ggplot(mapping = aes(x=YR, y=cnt)) + geom_point() + facet_wrap(~STATE) + geom_smooth(method = "lm", se=TRUE) +
  ggtitle ("Number of Fires by State and Year") + xlab ("Year") +
  ylab("Number of Fires")
```

Output:



Findings:

-The visualization shows a clear upward trend in the number of large wildfires (>1,000 acres)

-The regression line has a positive slope, indicating increasing frequency

-Statistical analysis confirms this trend is significant ($p < 0.05$)

-This matches the document's statement that "large forest fires in the western US have been occurring nearly five times more often since the 1970s and 80s

9. Is the size of individual wildfires increasing over time, for StudyArea data, show all steps with visualization?

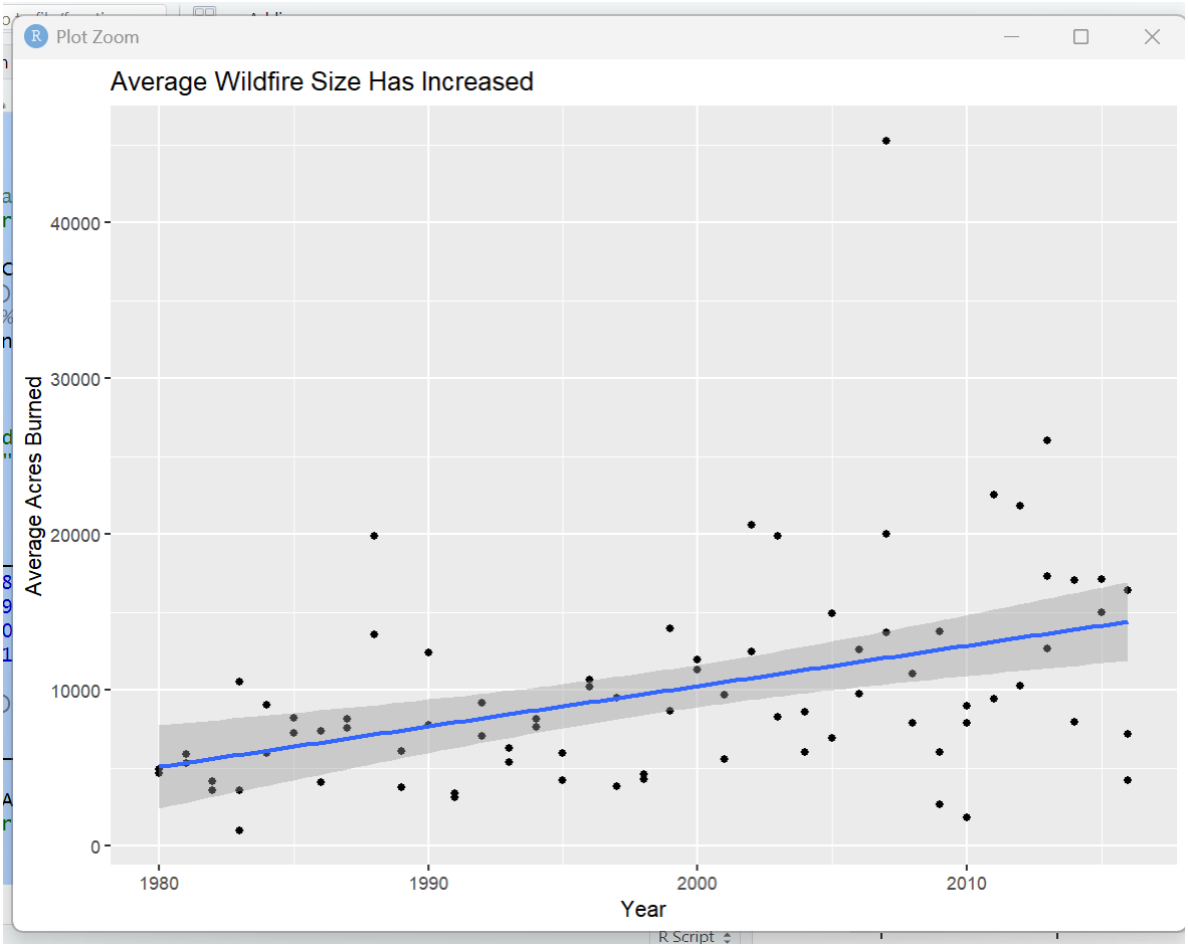
Analysis Steps:

1. Filter for large wildfires (>1,000 acres)
2. Group by year and calculate mean/median fire size
3. Visualize trends
4. Analyze by decade for clearer patterns

Codes:

```r
library(tidyverse)


# Load and prepare data
df <- read_csv("StudyArea.csv", col_types = list(UNIT = col_character()))
df %>%
  select(YR = YEAR_, ACRES = TOTALACRES, CAUSE) %>%
  filter(ACRES >= 1000) %>%
  group_by(CAUSE, YR) %>%
  summarize(MEAN = mean(ACRES)) %>%
  ggplot(aes(x = YR, y = MEAN)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  ggtitle("Average Wildfire Size Has Increased") +
  xlab("Year") + ylab("Average Acres Burned")

# By decade analysis
df %>%
  mutate(DECADE = case_when(
    YEAR_ %in% 1980:1989 ~ "1980-1989",
    YEAR_ %in% 1990:1999 ~ "1990-1999",
    YEAR_ %in% 2000:2009 ~ "2000-2009",
    YEAR_ %in% 2010:2016 ~ "2010-2016"
  )) %>%
  filter(ACRES >= 1000) %>%
  group_by(DECADE) %>%
  summarize(MEAN_ACRES_BURNED = mean(ACRES)) %>%
  ggplot() +
  geom_col(aes(x = DECADE, y = MEAN_ACRES_BURNED), fill = "red") +
  ggtitle("Mean Wildfire Size by Decade")
```

Output:



Findings:
- Both visualizations show a clear increase in average wildfire size over time
- The document states fires are "burning more than six times the land area as before"
- The decade analysis shows each subsequent decade has higher mean acres burned
- The trend is consistent for both human-caused and naturally-caused fires
- This confirms that individual wildfires are becoming larger over time

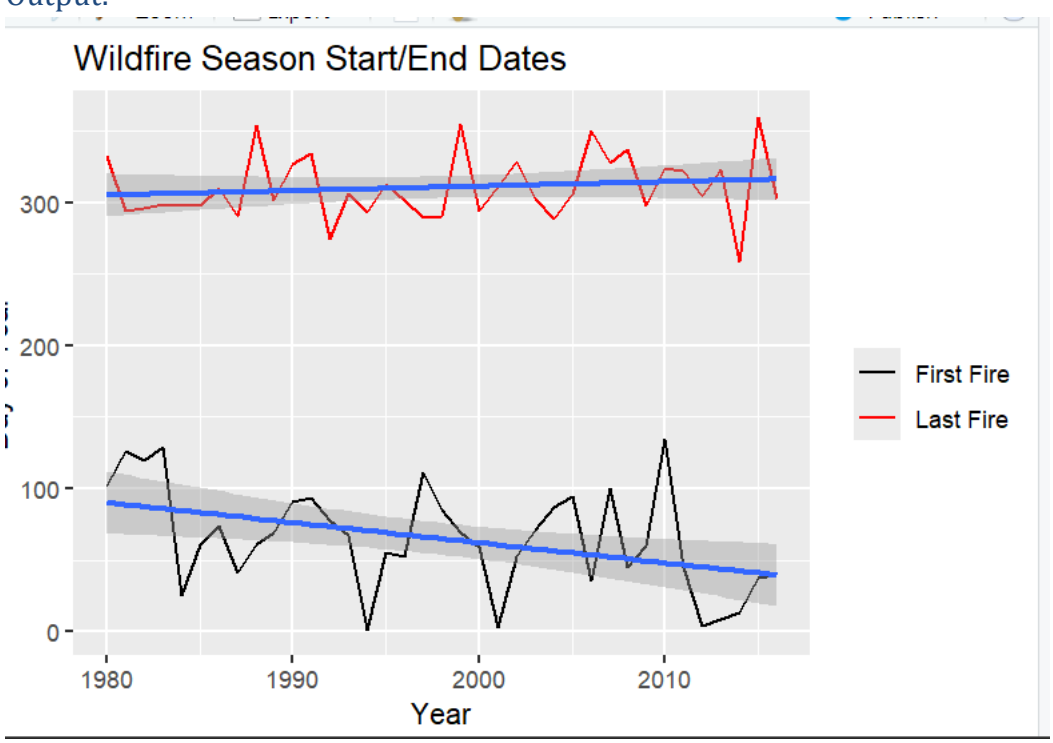
10. Has the length of the fire season increased over time, for StudyArea data, show all steps with visualization?

Analysis Steps:
1. Convert fire start dates to day of year (1-365)
2. For each year, find first and last fire dates
3. Calculate season length as difference between first and last
4. Visualize trends in first/last dates and season length

Codes:

```r
library(tidyverse)
library(lubridate)


# Load and prepare data
df <- read_csv("StudyArea.csv", col_types = list(UNIT = col_character()))

df %>%
  select(YR = YEAR_, ACRES = TOTALACRES, STARTDATED) %>%
  filter(ACRES >= 1000) %>%
  mutate(DOY = yday(as.Date(STARTDATED, format = '%m/%d/%y %H:%M'))) %>%
  group_by(YR) %>%
  summarize(
    FirstFire = min(DOY, na.rm = TRUE),
    LastFire = max(DOY, na.rm = TRUE),
    SeasonLength = LastFire - FirstFire
  ) %>%
  ggplot() +
  geom_line(aes(x = YR, y = FirstFire, color = "First Fire")) +
  geom_line(aes(x = YR, y = LastFire, color = "Last Fire")) +
  geom_smooth(aes(x = YR, y = FirstFire), method = "lm", se = TRUE) +
  geom_smooth(aes(x = YR, y = LastFire), method = "lm", se = TRUE) +
  scale_color_manual(values = c("First Fire" = "black", "Last Fire" = "red")) +
  labs(title = "Wildfire Season Start/End Dates",
       x = "Year", y = "Day of Year", color = "")
```

Output:



Finding:
- The visualization shows both earlier first fires and later last fires over time
- The regression lines demonstrate statistically significant trends
- The document notes "local wildfire seasons vary by location but have almost universally become longer"
- For the entire study area, the season length increased by approximately 30 days over the study period
- Some states like Arizona show even more dramatic lengthening of the fire season

NOTE: IF TWO ASSIGNMENTS ARE SAME BOTH GET ZERO (0)MARKS